

Multiple Choice (10 marks)

1. Neural networks:
 - (a) Optimize a convex objective function
 - (b) Can only be trained with stochastic gradient descent
 - (c) Can use a mix of different activation functions
 - (d) None of the above
2. What is the dimensionality of the null space of the following matrix? $A = \begin{bmatrix} 3 & 2 & -9 \\ -6 & -4 & 18 \\ 12 & 8 & -36 \end{bmatrix}$
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
3. What is the rank of the following matrix? $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
4. Which of the following best describes the joint probability distribution $P(X, Y, Z)$ for the given Bayes net. $X \leftarrow Y \rightarrow Z$?
 - (a) $P(X, Y, Z) = P(Y) * P(X|Y) * P(Z|Y)$
 - (b) $P(X, Y, Z) = P(X) * P(Y|X) * P(Z|Y)$
 - (c) $P(X, Y, Z) = P(Z) * P(X|Z) * P(Y|Z)$
 - (d) $P(X, Y, Z) = P(X) * P(Y) * P(Z)$
5. Statement 1| The Stanford Sentiment Treebank contained movie reviews, not book reviews. Statement 2| The Penn Treebank has been used for language modeling.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: A small step size can lead to an increase in training loss due to insufficient model capacity to learn.
7. True or False: The polynomial degree is a critical factor affecting the trade-off between underfitting and overfitting in polynomial regression.
8. True or False: MLE estimates are typically unbiased and consistently converge to the true parameter value as sample size increases.
9. True or False: Both the original ResNets and Transformers are feedforward neural networks.
10. True or False: If $P(A, B)$ decreases while $P(A)$ increases, then $P(B)$ must also decrease.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the triplet with sum of the given array
12. Write a function to repeat the given tuple n times.

End of Paper